



ROBERTSON

Tech Training Session 3

DA, CYBR, FSD
March 25, 2025

Land Acknowledgement

At Robertson College we acknowledge that we work and study:

In Winnipeg on Treaty 1 territory, the traditional gathering place of the Anishinaabe, Cree, Oji-Cree, Dakota and Dene people and the traditional homeland of the Métis people.

In Edmonton on Treaty 6 territory, a traditional gathering place, travelling route and home for many Indigenous Peoples including the Nehiyawak/Cree, Tsuut'ina, Niitsitapi/Blackfoot, Métis, Nakota Sioux, Haudenosaunee/Iroquois, Dene Suliné, Anishinaabe/Ojibway/Saulteaux, and the Inuk/Inuit.

In Calgary on the traditional territories of the Blackfoot Confederacy (Siksika, Kainai, Piikani), the Tsuut'ina, the Îyâxe Nakoda Nations, the Métis Nation (Region 3), and all people who make their homes in the Treaty 7 region of Southern Alberta.

Robertson's Vision

Robertson is the School of New Work.
Our vision is for everyone to have a good life
through a prosperous and fulfilling career.



Robertson's Mission

Our mission is to empower our students with the most relevant training and up-to-date industry insights to thrive in a modern work environment.

We are guides for our students during their time at Robertson.



Agenda

- Tech Session 1 Review
- Session Objectives
- Program Performance
- Program Voucher - Review
- Program Readiness and Fit
- What to Expect During Practicum and Initial Employment
- Need to Know Information
- Labour Market Trends, Gaps and Needs
- Discussion/Q&A

Session Objectives



The purpose of this sessions is to:

- Review program-specific performance metrics.
- Discuss labour market trends, gaps and needs.
- Discuss specific program details, including student fit/readiness and what to expect during practicum and initial employment.
- Explore program-specific information provided in the Program Knowledge Course in Brightspace.

Tech Program Performance



Program	Enrolment	Student Satisfaction	Graduation (TTM)	Employment
DA	15%	4.25	91%	31%
CYBR	30%	4.29	83%	35%
FSD	30%	4.62	50%	50%
Data Source	<i>Reflects the number of students enrolment compared to enrolment capacity (seats)</i> *See data source *See data source	*Mean score out 5 **Retrieved from Course Evaluation Surveys	Retrieved from ITM Graduation Report	*Reflects TTM Advertised Rate Retrieved from ITM Adv Emp Report

Tech Certifications and Voucher Overview



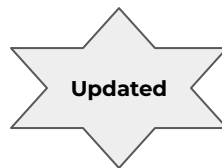
Program	Vouchers Offered	Approximate Value	Vouchers Requested Jan. 1 - Dec. 31, 2024
DA	Oracle Database SQL Microsoft Azure Fundamentals	\$490	N/A
CYBR	Systems Security Certified Practitioner (SSCP) Certificate of Cloud Security Knowledge (CCSK) Certified Azure Administrator Associate AWS Certified Architect Associate	\$1150 CAD	SSCP: 5 Azure: 4
FSD	Java SE 11 Developer Oracle Database SQL Microsoft Azure Fundamentals	\$840 CAD	Oracle: 1

← **ADDED**

Dollar value for each program added to the website.

- Robertson College strongly encourages all students in the School of Technology to challenge all vendor exams associated with their program to obtain vendor-based certificates/diplomas.
- These additional certifications will complement the Robertson College diploma and are sought after by industry.
- Vouchers can be requested while students are in the program and up to 18 months after their graduation date.
- Voucher requests have been consistently low Y-O-Y.

***See [Tech Certification Guide](#) for comprehensive overview.**



Quick Poll



Students are required to request vouchers for all industry relevant certification exams that are offered.

YES



NO



Quick Poll

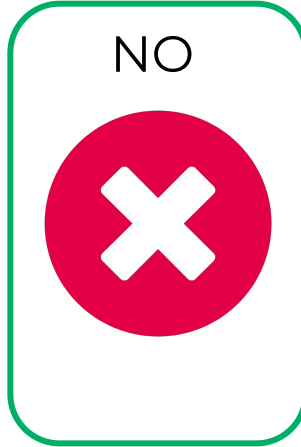


Students are required to request vouchers for all industry relevant certification exams that are offered.

YES



NO



DA: Labour Market Trends, Gaps and Needs

	Region	2025 Jobs	2030 Jobs	Change	% Change
●	2 Provinces	8,461	9,255	794	9.4%
●	National Average	10,845	11,791	946	8.7%
●	Manitoba	975	1,002	27	2.8%
●	Alberta	7,486	8,253	767	10.3%

The number of jobs in this role will increase by the percentages above between 2025-2030 (Lightcast, OO, NOC 21223, 2025).

See [DA: Data Analyst](#) for more information.

Labour market demand over the next three years:
Moderate in AB
Very Good in MB.
(Government of Canada, NOC 21223, 2025).

Between Jan, 2024 and Feb, 2025 there were 2,719 unique job postings for this role: 2311 in Alberta and 408 MB (Lightcast, JPA, NOC 21223, 2025).

Top companies posting for similar roles include Si Systems, Government of Alberta and public universities (Lightcast, JPA, NOC 21223, 2025).

Full Stack: Labour Market Trends, Gaps and Needs

	Region	2025 Jobs	2030 Jobs	Change	% Change
●	2 Provinces	3,488	3,689	201	5.8%
●	National Average	4,755	4,915	160	3.4%
●	Manitoba	758	802	44	5.8%
●	Alberta	2,730	2,887	157	5.8%

The number of jobs in this role will increase by the percentages above between 2025-2030 (Lightcast, OO, NOC 21234 , 2025).

See [FSD: Full Stack Web Developer](#) for more information.

Labour market demand over the next three years:
Moderate in AB
Good in MB.
(Government of Canada, NOC 21234, 2025).

Between Jan, 2024 and Feb, 2025 there were 660 unique job postings for this role: 596 in Alberta and 64 in MB (Lightcast, JPA, 21234, 2025).

Top companies posting for similar roles include Si Systems, Smart--Technologies, Hootsuite Media (Lightcast, JPA, 21234, 2025).

Cyber: Labour Market Trends, Gaps and Needs

	Region	2025 Jobs	2030 Jobs	Change	% Change
●	2 Provinces	2,105	2,303	198	9.4%
●	National Average	3,217	3,511	294	9.1%
●	Manitoba	454	504	50	11.0%
●	Alberta	1,652	1,800	148	9.0%

The number of jobs in this role will increase by the percentages above between 2025-2030 (Lightcast, OO, NOC 21220, 2025).

See [CYBR: Cyber and Cloud Security Professional](#) for more information.

Labour market demand over the next three years:
Good in AB
Good in MB.
(Government of Canada, NOC 21220, 2025).

Between Jan, 2024 and Feb, 2025 there were 853 unique job postings for this role: 711 in Alberta and 142 in MB (Lightcast, JPA, NOC 21220, 2025).

Top companies posting for similar roles include Si Systems, Long View Systems, Canada Life and Procom (Lightcast, JPA, NOC 21220, 2025).

Introduction to Instructors/SMEs



Jesia Yuki
DA



Joanne Hoar
FSD



Christopher Nelson
Cyber and Cloud

Jesia Yuki: DA



Professional Background

- 5+ years of experience
- Master's in Data Science
- Coursera Data Analyst Certification
- Skilled in SQL, Python, R, Tableau, and Power BI.



Explanation of DA Roles

- Collect, clean, and interpret data to uncover insights that drive business decisions.
- Use tools like SQL, Python, and Tableau, to identify trends, create visualizations, and generate reports.



What you are seeing from students

- Students begin with minimal coding skills but quickly gain proficiency.
- They develop expertise in data cleaning, visualization, and statistical analysis.
- Hands-on projects and real-world case studies help refine problem-solving skills.



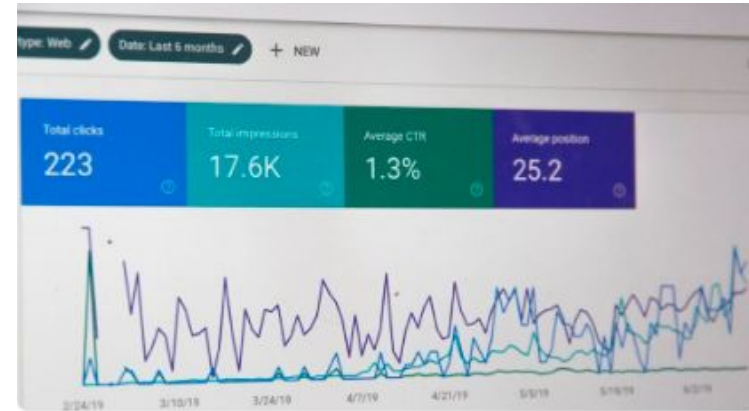
What prospective students need to know

- Certification can provide a structured learning path and industry-recognized credentials to enhance career opportunities.
- A growth mindset and curiosity about patterns in data will set students apart in this competitive field.

Key Needs in the Tech Sector for Data Analysts



- **Technical Expertise:** Proficiency in SQL, Python, R, cloud platforms, and BI tools for data processing and visualization.
- **Market Demands:** Businesses need versatile analysts skilled in marketing and operations to drive informed decisions, optimize processes, and stay competitive in a data-driven world.
- **Strategic Contribution:** Analysts transform complex data into real-time, actionable insights, helping tech companies improve user experiences, enhance security, and scale operations..



Source: Photo by Isaac Smith on UnSplash

Identifying Gaps in Data Analyst Skills



- **Skill Shortages:** There is a high demand for data analysts, but many job candidates lack key skills, limiting effective data use.
- **Education Gaps:** Traditional educational frameworks struggle to keep pace with the rapid advancements in data technologies, resulting in a disconnect between academic curricula and industry needs.
- **Industry Expectations:** Employers want experienced analysts, but many new hires lack the hands-on skills needed for the job.

Student Readiness and Fit:



Student Readiness for Data Analyst Program

-  **Data Tools:** Have familiarity with basic skills such as Excel, SQL and Python.
-  **Analytical Skills:** Able to analyze data and identify patterns.
-  **Data Ethics:** Understanding of data privacy, security, and compliance regulations.
-  **Tech Adaptability:** Eager to learn AI, machine learning, and cloud tools.
-  **Certifications:** Interest in earning certifications like Google Data Analytics, Microsoft Certified Data Analyst, or AWS Data Analytics.

Student Fit for Data Analyst Program

-  **Passion for Data:** Enthusiastic about using data for insights.
-  **Hands-On:** Enjoys working with real data and solving problems.
-  **Career-Focused:** Aiming for roles like Data Analyst or Business Analyst.
-  **Detail-Oriented:** Critical thinker with strong troubleshooting skills.
-  **Growth Mindset:** Committed to learning and staying current in analytics.

What Students Should Expect on Practicum and Initial Employment

- **Work with Real Data:** Clean, transform, and analyze datasets using tools like SQL, Excel, Python, R, Tableau, and Power BI.
- **Solve Business Problems:** Analyze data to find trends, generate insights, and create visualizations.
- **Collaborate with Teams:** Work with mentors, cross-functional teams, and stakeholders to solve data-driven problems.
- **Use Analytical Tools:** Apply industry-standard tools for data analysis, visualization, and reporting.
- **Receive Feedback:** Get feedback from mentors and team members to improve analysis and presentation skills.
- **Learn Data Governance:** Understand data privacy, security, and compliance regulations (e.g., GDPR, HIPAA).
- **Report Insights:** Prepare reports and dashboards to communicate findings to business leaders.
- **Continuous Learning:** Stay updated on new tools, trends, and techniques in data analysis and business intelligence.
- **Handle Deadlines:** Work under pressure to meet deadlines for reports, analysis, and project milestones.

Prospective/Current/Graduates Need to Know that....

Prospective and current data analyst students: they should prioritize continuous learning, hands-on experience, certifications, critical thinking, and ethical data handling to build a successful career.

Current students: they need to enhance practical skills, stay current with industry tools and trends, earn certifications, and strengthen problem-solving and communication abilities to excel in the field.

Graduate data analyst students: they must refine their expertise, gain real-world experience, pursue advanced certifications, and stay on top of evolving technologies and data trends.

Joanne Hoar: FSD



Professional Background

- 20 years in game dev, scientific modelling and web tech consulting
- M.Sc. CPSC
- Cloud code back-end services



Explanation of FSD Roles

- Building comprehensive web solutions
- Concept to product delivery to ongoing maintenance
- “Jack/Jill of all trades”



What you are seeing from students

- Comprehensive = versatile
- Exposure to tools = career readiness
- Pace too fast can be overwhelming
- Keeping up with trends
- Real-world debugging tricky



What prospective students need to know

- Involves both front and back end
- Requires basic web concepts
- Significant time investment
- Building a portfolio
- Career goals

Student Readiness and Fit: FSD



Readiness:

- Good organizational skills and communication skills
- Prepared for a fast-paced, challenging career
- Understand components of computer systems and networking
- Understand documentation and able to read code
- Web fundamentals is helpful

Fit:

- Able to learn on-the-fly (life long learner)
- Analytical, big picture thinking
- Methodical and rigorous, systematic and exacting
- Team player

What Students Should Expect on Practicum and Initial Employment (FSD)

Key member of multidisciplinary team including developers, business analysts, project managers, and product owners

May participate at any or all level of the stack eg. applications, databases, web, test automation, requirements gathering, etc

- Fast-paced, steep learning curve
- Each company has its own tech stack
- Each company has its own security, accessibility and usability guidelines
- Industry-wide guidelines change rapidly

Prospective/Current/Graduates Need to Know that:

- They should master the fundamentals (since the specifics will change)
- It's important to embrace both frontend and backend
- They will write clean, maintainable code (and technical documentation)
- Soft skills will matter (communication, collaboration, time management)
- They should join communities and networks
- Building and maintaining a portfolio is important for career attainment and advancement.

Christopher Nelson: CYBR



Professional Background

- 20 years of experience
- ISC2 Global Trainer
- Empower students at all levels with the knowledge and skills needed to excel in the evolving cybersecurity landscape.



Explanation of Cyber and Cloud Roles

- Protect digital assets by securing networks, systems, and cloud environments.
- Implement security measures, monitor threats, and ensure data protection.
- Defend against cyber risks and enhance overall security resilience.



What you are seeing from students

- Students seek expertise in high-demand areas like network and application security, IAM, threat intelligence, incident response, and penetration testing.



What prospective students need to know

- This industry requires continuous learning.
- Skills in risk management, encryption, identity management, and compliance is key.

Student Readiness and Fit for Cybersecurity and Cloud:



Student Readiness

- Basic IT, networking, and security knowledge
- Strong problem-solving and analytical skills
- Ethical mindset and understanding of compliance
- Adaptability to evolving technologies
- Interest in industry certifications (e.g., SSCP, CISSP, CompTIA Security+, AWS Cloud Practitioner)

Student Fit

- Passion for cybersecurity and cloud computing
- Hands-on learning and technical curiosity
- Career goals in security, cloud, or IT roles
- Ability to work under pressure and think critically
- Commitment to continuous learning and professional growth

What Students Should Expect on Practicum and Initial Employment (CYBR)

- Hands-on experience with security tools and cloud platforms
- Real-world threat analysis and risk assessments
- Incident response and troubleshooting tasks
- Collaboration with IT and security teams
- Learning industry best practices and compliance standards
- Fast-paced, problem-solving environment with continuous learning opportunities
- Find and fix bugs
- Execute test cases
- Coordinate with developers
- Collaborate with other departments
- Report on testing

Prospective/Current/Graduates Need to Know that....

- Prospective and current cybersecurity students must embrace continuous learning, hands-on experience, certifications, problem-solving, and ethical responsibility for career success.
- Current cybersecurity students need to develop hands-on skills, stay updated on threats, earn certifications, and build problem-solving and ethical decision-making abilities.
- Graduate cybersecurity students must refine expertise, gain industry experience, pursue advanced certifications, and stay updated on evolving threats and technologies.

Program Knowledge Training Course

[Data Analyst](#)

[Full Stack Web Developer](#)

[Cyber and Cloud Security](#)

[Professional](#)

Program Info and Video

Program Outline

Job Profile

Labour Market Information

Quick Poll



Students are highly satisfied with the DA, CYBR and FSD programs.

YES



NO



Quick Poll



Students are highly satisfied with the DA, CYBR and FSD programs.



Quick Poll



Robertson recently added 2 vouchers to help students obtain certification as data analysts.

YES



NO



Quick Poll



Robertson recently added 2 vouchers to help students obtain certification as data analysts.



Quick Poll



In addition to technical skills, students in tech programs should also have or develop the following skills/aptitudes:

- A. Communication
- B. Collaboration
- C. Growth mindset
- D. All of the above

Quick Poll



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- D. All of the above

Quick Poll



Where can I find a short intro video that explains the typical work day and tips for success for each tech program?

- A. Robertson Academy - Onboarding Course
- B. Robertson Academy - Program Knowledge Training Course
- C. Robertson Academy - Instructor Training Course
- D. Robertson Academy - Policies Course

Quick Poll



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- D. Robertson Academy - Policies Course

Q and A

Discussion/Q&A

1. What are some key lessons you have learned as a professional in Tech that you would like to share?
2. How can we support new Tech Students to be “more ready” for their program prior to start? What advice would you (instructors/SMEs) give to prospective students about the tech industry?
3. How can we further communicate/emphasize the value of tech vouchers and certifications to students?
4. What is prohibiting enrolment into Robertson Tech programs? How can we increase it?
5. What is prohibiting employment of Robertson Tech grads? How can we increase it?

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